Public-Key Cryptosystems from the Worst-Case Shortest Vector Problem

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Abstract: Public-Key Cryptosystems from LWE

- **Objective:** Construction of public-key cryptosystems secure under the **worst-case hardness** of approximating the minimum distance of n-dimensional lattices to within small poly(n) factors.

Main Technical Innovation

A novel reduction from variants of the **Shortest Vector Problem (SVP)** to corresponding versions of the **Learning with Errors (LWE)** problem.

- **Context:** Previously, only quantum reductions of this kind were known.
- **This Work:** Provides a classical reduction connection.

Additional Contributions

- Construction of a **Chosen Ciphertext-Secure (CCA)** cryptosystem.
- Simpler description and tighter underlying worst-case approximation factors than prior lattice-based schemes.

References: Ajtai-Dwork (1997), Regev (2004, 2005)

Core Results: Bridging GapSVP and LWE

The Main Breakthrough

The first public-key cryptosystem with security based on the **worst-case hardness** of approximating the Shortest Vector Problem (GapSVP) on arbitrary lattices via a **classical reduction**.

- **Innovation:** A purely classical (probabilistic polynomial-time) reduction from lattice problems to Learning With Errors (LWE).
- **Hardness Result:** For large moduli $q \geq 2^{n/2}$:

$$\text{Search-LWE} \geq \text{GapSVP}_{\tilde{O}(n/\alpha)}$$

- **Noise Parameter:** $\alpha \cdot q$ represents the standard deviation of the Gaussian error over $\mathbb{Z}_q$.

A Novel Lattice Problem variant

Moduli as small as $q \geq n/\alpha$

The reduction holds for smaller q by introducing a novel variant: **ζ-to-γ-GapSVP**.

Definition: Approximate the minimum distance to within γ, given a promise that it lies in a range with gap $\zeta > \gamma$.

Properties:

- Equivalent to standard GapSVP for $\zeta \geq 2^{n/2}$.
- For $\zeta = \text{poly}(n)$, it remains **exponentially hard** in dimension n.

The Benefit

Relying on this variant allows for a **smaller choice of** q, making the resulting cryptosystems more efficient.

Applying Hardness to Cryptography

- **Prior Schemes:** Public-key (Reg05, PVW08) and Identity-Based Encryption (GPV08) rely on **Decision-LWE**.
- **The Search-to-Decision Bridge:**
  - For prime modulus $q = \text{poly}(n)$, Search $\equiv$ Decision.
  - For Gaussian errors, Search $\equiv$ Decision when q is a product of many distinct small primes.

Final Impact

By adapting prior systems to these specific values of q, we obtain **semantically secure** cryptosystems based on the worst-case hardness of:

- Standard **GapSVP** (with large q)
- The ζ-**to-**γ-**GapSVP** variant (with smaller q)

Background: Prior Cryptosystems

The Common Approach

Most cryptosystems with worst-case connections (e.g., AD97, Reg04) rely on distinguishing between two types of probability distributions over a domain:

- **Uniform Distribution:** Evenly spread across the domain.
- **Lumpy Distribution:** Highly concentrated over specific parts of the domain.

- **The Core Reduction:** Solving a worst-case lattice problem $\rightarrow$ Distinguishing between uniform and lumpy distributions.
- **Geometric Constraints:** Historically, the input must obey constraints during the reduction (e.g., the lattice must have a **"unique" shortest vector**).

Regev's Iterative Reduction [Reg05]

Regev's reduction consists of two alternating components that feed back into each other:

1. Classical Component

Given an LWE oracle and specific lattice points, it solves **Bounded-Distance Decoding (BDD)** within a certain radius.

2. Quantum Component

Uses a BDD oracle to sample lattice points from a more concentrated distribution.

The Feedback Loop: Samples from the quantum step are fed back to the classical step to solve BDD for an *increasingly larger* decoding radius.

Classical vs. Quantum Limitations

The Main Obstacle to Classical Reductions

How do we generate useful inputs for a BDD oracle?

- **The Classical Paradox:** If we choose a lattice point $\mathbf{y}$ and add a small perturbation $\mathbf{z}$ to get $\mathbf{x} = \mathbf{y} + \mathbf{z}$, the oracle outputs $\mathbf{y}$.
- *"Clearly, on input* $\mathbf{x}$ *the oracle outputs* $\mathbf{y}$*. But this is useless since we already know* $\mathbf{y}$*!"* [Reg05]

The Quantum Advantage

Regev's quantum reduction uses the BDD oracle to **uncompute y** from **x**. This ability to operate on superpositions of points turns a "useless" identity into a powerful sampling tool.

Probability: Statistical Distance

Statistical Distance (Total Variation Distance) The statistical distance between two distributions X and Y over D is defined as:

$$\Delta(X, Y) = \max_{A \subseteq D} |f_X(A) - f_Y(A)|$$

Key Properties

- **Metric:** It obeys the triangle inequality.
- **Data Processing Inequality:** For any (possibly randomized) function g:

$$\Delta(g(X), g(Y)) \leq \Delta(X, Y)$$

Notation: The uniform distribution over a domain D is denoted as $U(D)$.

The n-dimensional Gaussian Distribution

For $n \in \mathbb{Z}^+$ and parameter $r > 0$, the **Gaussian function** $\rho_r^{(n)} : \mathbb{R}^n \to \mathbb{R}$ is:

$$\rho_r^{(n)}(\mathbf{x}) = \exp(-\pi(\|\mathbf{x}\|/r)^2)$$

- **Total Measure:** $\int_{\mathbb{R}^n} \rho_r(\mathbf{x})\, d\mathbf{x} = r^n$.
- **Density Function:** $D_r(\mathbf{x}) = \rho_r(\mathbf{x})/r^n = D_r^{(n)}(\mathbf{x})$.

Tail Inequalities

The distribution is spherically symmetric. For $\mathbf{x} \sim D_r^{(n)}$:

- **1D Case:** $P(|x| \geq r \cdot t) \leq \exp(-\pi t^2)$ for $t \geq 1$.
- **nD Case:** $P(\|\mathbf{x}\| \geq r\sqrt{n}) \leq 2^{-n}$.

Note: $r = \sigma\sqrt{2\pi}$, Here σ is a S.D. on $D_r^{(1)}$

Sampling and Geometric Lemmas

Sampling Efficiency

- It is possible to sample efficiently from D_r and $U(\mathcal{B}_n)$ (uniform over an n-dimensional ball).
- In practice, calculations use a polynomial number of bits of precision.

Lemma [GG00] : Overlapping Balls

For constants $c, d > 0$ and any $\mathbf{z} \in \mathbb{R}^n$ with $\|\mathbf{z}\| \leq d$ and $d' = d \cdot \sqrt{n/(c \log n)}$, we have:

$$\Delta(U(d' \cdot \mathcal{B}_n), U(\mathbf{z} + d' \cdot \mathcal{B}_n)) \leq 1 - 1/\text{poly}(n)$$

Interpretation: If the centers of two n-dimensional balls are relatively close, their statistical distance is bounded away from 1.

Lattices: Definitions and Properties

Definition: n-dimensional Lattice

- A discrete additive subgroup of $\mathbb{R}^n$.
- Generated by a basis $\mathbf{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\} \subset \mathbb{R}^n$:

$$\Lambda = \mathcal{L}(\mathbf{B}) = \left\{ \mathbf{Bz} = \sum_{i \in [n]} z_i \cdot \mathbf{b}_i : \mathbf{z} \in \mathbb{Z}^n \right\}$$

Fundamental Parallelepiped

The origin-centered region $\mathcal{P}_{1/2}(\mathbf{B})$ is defined as:

$$\mathcal{P}_{1/2}(\mathbf{B}) = \left\{ \mathbf{Bc} = \sum_{i \in [n]} c_i \cdot \mathbf{b}_i : \mathbf{c} \in [-\tfrac{1}{2}, \tfrac{1}{2})^n \right\} \subset \mathbb{R}^n$$

Modular Reduction

For $\mathbf{w} \in \mathbb{R}^n$, $\mathbf{x} = \mathbf{w} \bmod \mathbf{B}$ is the unique $\mathbf{x} \in \mathcal{P}_{1/2}(\mathbf{B})$ such that $\mathbf{w} - \mathbf{x} \in \mathcal{L}(\mathbf{B})$. Computed as: $\mathbf{x} = \mathbf{B}(\mathbf{B}^{-1}\mathbf{w} - \lfloor \mathbf{B}^{-1}\mathbf{w} \rceil)$.

Minimum Distance and Dual Lattices

Minimum Distance $\lambda_1(\Lambda)$ The length of the shortest nonzero lattice vector (typically in ℓ_2 norm):

$$\lambda_1(\Lambda) = \min_{\mathbf{x} \neq \mathbf{y} \in \Lambda} \|\mathbf{x} - \mathbf{y}\| = \min_{0 \neq \mathbf{x} \in \Lambda} \|\mathbf{x}\|$$

Lower bound: $\lambda_1(\Lambda) \geq \min_i \|\widetilde{\mathbf{b}}_i\|$, where $\widetilde{\mathbf{b}}_i$ are Gram-Schmidt vectors.

Dual Lattice Λ^* Defined as the set of points whose inner product with any lattice vector is an integer:

$$\Lambda^* = \{\mathbf{x} \in \mathbb{R}^n : \forall \mathbf{v} \in \Lambda, \langle \mathbf{x}, \mathbf{v} \rangle \in \mathbb{Z}\}$$

- **Symmetry:** $(\Lambda^*)^* = \Lambda$.
- **Dual Basis:** If $\mathbf{B}$ is a basis for Λ, then $\mathbf{B}^* = (\mathbf{B}^{-1})^t$ is a basis for Λ^*.

Dual Basis and Gram-Schmidt Orthogonalization

Lemma: Dual Gram-Schmidt Vectors

Let $\mathbf{B}$ be an ordered basis and $\mathbf{D} = \mathbf{B}^* = \mathbf{B}^{-t}$ be its dual basis. Let $\widetilde{\mathbf{b}}_i$ and $\widetilde{\mathbf{d}}_i$ be their respective Gram-Schmidt orthogonalizations. Then:

$$\widetilde{\mathbf{d}}_i = \frac{\widetilde{\mathbf{b}}_i}{\|\widetilde{\mathbf{b}}_i\|^2} \quad \text{for all } i \in [n]$$

In particular, it follows that $\|\widetilde{\mathbf{d}}_i\| = 1/\|\widetilde{\mathbf{b}}_i\|$.

Proof Sketch:

- Write $\mathbf{B} = \mathbf{QU}$ ($\mathbf{U}$ is upper triangular).
- Then $\mathbf{D} = \mathbf{B}^{-t} = \mathbf{Q} \cdot \mathbf{U}^{-t}$.
- Since $\mathbf{U}^{-t}$ is lower triangular, this forms the Gram-Schmidt decomposition of the dual basis in **reverse order**.
- $\widetilde{\mathbf{b}}_i = q_i.u_{ii}$ AND $\widetilde{\mathbf{d}}_i = q_i.u_{ii}^{-1}$
- $\mathbf{Q}$ is orthogonal.

Decoding Algorithms: Babai's Rounding

To secure a cryptosystem, we use decoding algorithms to find lattice points near a vector $\mathbf{x} \in \mathbb{R}^n$.

1. **Simple Rounding Algorithm** Given a basis $\mathbf{B}$ and a target point $\mathbf{x}$:

$$\mathbf{v} = \mathbf{B} \cdot \lfloor \mathbf{B}^{-1}\mathbf{x} \rceil \in \mathcal{L}(\mathbf{B})$$

- **Decision Region:** The set of points $\mathbf{x}$ that map to a specific lattice point $\mathbf{v}$ is the parallelepiped:

$$\mathbf{x} \in \mathbf{v} + \mathcal{P}_{1/2}(\mathbf{B})$$

Decoding Algorithms: Nearest-Plane

2. Babai's Nearest-Plane Algorithm A more powerful iterative decoding approach. Given basis $\mathbf{B}$ and point $\mathbf{x}$:

The Algorithm

1. Let $\mathbf{y} \leftarrow \mathbf{x}$.
2. For $i = n, \ldots, 1$:
   - $c_i = \lfloor \langle \mathbf{y}, \widetilde{\mathbf{b}}_i \rangle / \langle \widetilde{\mathbf{b}}_i, \widetilde{\mathbf{b}}_i \rangle \rceil$
   - $\mathbf{y} \leftarrow \mathbf{y} - c_i \mathbf{b}_i$
3. **Output:** $\mathbf{x} - \mathbf{y} \in \mathcal{L}(\mathbf{B})$ *and* $\mathbf{y} \in \mathcal{P}_{1/2}(\widetilde{\mathbf{B}})$

- **Decision Region:** This algorithm outputs $\mathbf{v} \in \mathcal{L}(\mathbf{B})$ for any point in the region:

$$\mathbf{v} + \mathcal{P}_{1/2}(\widetilde{\mathbf{B}})$$

where $\widetilde{\mathbf{B}}$ represents the **orthogonalized** basis.

Shortest Vector Problem (GapSVP)

Definition: GapSVP_γ

For an approximation factor $\gamma(n) \geq 1$, the input is a pair $(\mathbf{B}, d)$.

- **YES instance:** $\lambda_1(\Lambda) \leq d$
- **NO instance:** $\lambda_1(\Lambda) > \gamma(n) \cdot d$

Approximation via Oracle

- Given an oracle for GapSVP_γ, we can approximate the minimum distance $\lambda_1(\Lambda)$ within a factor of 2γ.
- This is achieved efficiently using a **binary search** on the value d.

The Generalized ζ-to-γ-GapSVP

Definition : Generalized GapSVP A more specific version used in worst-to-average-case reductions. Input is $(\mathbf{B}, d)$ where:

1. $\lambda_1(\Lambda) \leq \zeta(n)$
2. $\min_i \|\widetilde{\mathbf{b}}_i\| \geq 1$
3. $1 \leq d \leq \zeta(n)/\gamma(n)$

Decision Criteria

- **YES:** $\lambda_1(\Lambda) \leq d$
- **NO:** $\lambda_1(\Lambda) > \gamma(n) \cdot d$

Note: This provides a "promise" that the shortest vector lies within a looser range $\zeta(n)$ while asking for a tighter gap $\gamma(n)$.

Analysis of Conditions

Why the conditions matter:

- **Scaling:** Condition 2 ($\min_i \|\widetilde{\mathbf{b}}_i\| \geq 1$) can be achieved without loss of generality by scaling the basis $\mathbf{B}$.
- **Triviality:** Condition 3 ensures the problem isn't trivally solvable by d lying outside the effective range.

The LLL Algorithm and Large ζ

For $\zeta(n) \geq 2^{n/2}$, the problem is equivalent to standard GapSVP$_\gamma$. Using the **LLL algorithm**, we can reduce any basis such that:

$$\lambda_1(\Lambda) \leq \|\mathbf{b}_1\| \leq 2^{n/2} \cdot \min_i \|\widetilde{\mathbf{b}}_i\|$$

Hardness: For $\zeta(n) = \text{poly}(n)$, no known algorithms solve this in time better than $2^{\Omega(n)}$, even with a "high quality" basis promise.

The Smoothing Parameter

Micciancio and Regev [MR07] introduced the **smoothing parameter** $\eta_\epsilon(\Lambda)$, which relates the lattice structure to Gaussian measures.

Definition : Smoothing Parameter

For an n-dimensional lattice Λ and $\epsilon > 0$, $\eta_\epsilon(\Lambda)$ is the smallest r such that:

$$\rho_{1/r}(\Lambda^* \setminus \{\mathbf{0}\}) \leq \epsilon$$

- **Scaling Property:** For any $c \neq 0$, $\eta_\epsilon(c \cdot \Lambda) = c \cdot \eta_\epsilon(\Lambda)$.
- **General Bound [MR07, Lemma 3.2]:** $\eta_{2^{-n}}(\Lambda) \leq \sqrt{n}/\lambda_1(\Lambda^*)$.
- **Integer Lattice:** For $f(n) = \omega(\sqrt{\log n})$, there exists negligible ϵ such that $\eta_\epsilon(\mathbb{Z}) \leq f(n)$ [MR07].

Properties of the Smoothing Parameter

When the Gaussian parameter r exceeds $\eta_\epsilon(\Lambda)$, the lattice "looks" like a continuous space in terms of Gaussian measure.

Lemma [MR07, Lemma 4.4]

For any lattice Λ, $\epsilon \in (0, 1)$, $r \geq \eta_\epsilon(\Lambda)$, and any shift $\mathbf{c} \in \mathbb{R}^n$:

$$\rho_r(\Lambda + \mathbf{c}) \in \left[\frac{1-\epsilon}{1+\epsilon}, 1\right] \cdot \rho_r(\Lambda)$$

Significance:

- This implies that the Gaussian measure associated with any *translate* (coset) of the lattice is essentially unchanged.
- This "smoothness" is vital for the security proofs of LWE and other lattice primitives.

Discrete Gaussians and Sampling

Discrete Gaussian Distribution For a lattice Λ and $r > 0$, the distribution $D_{\Lambda,r}$ over Λ (centered at zero) is defined as:

$$D_{\Lambda,r}(\mathbf{x}) = \frac{\rho_r(\mathbf{x})}{\rho_r(\Lambda)} \quad \forall \mathbf{x} \in \Lambda$$

Proposition : Efficient Sampling [GPV08, Theorem 4.1]

There exists a probabilistic polynomial-time (PPT) algorithm that:

- **Input:** Basis $\mathbf{B}$ and $r \geq \max_i \|\widetilde{\mathbf{b}}_i\| \cdot \omega(\sqrt{\log n})$.
- **Output:** A sample within $\mathrm{negl}(n)$ statistical distance of $D_{\Lambda,r}$.

Note: This sampling algorithm (often called the GPV sampler) is a core subroutine in many lattice-based signatures and encryptions.

Learning with Errors: Mathematical Setting

The Group $\mathbb{T}$ Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ be the additive group on the real interval $[0, 1)$ with **modulo 1 addition**.

Parameters

- $n, q \geq 2$: Positive integers.
- $\mathbf{s} \in \mathbb{Z}_q^n$: A secret vector.
- ϕ: A probability distribution on $\mathbb{T}$ (the error distribution).

Gaussian Error on $\mathbb{T}$ For $\alpha > 0$, we define Ψ_α as the distribution obtained by:

1. Drawing a sample from the 1D Gaussian $D_\alpha^{(1)}$.
2. Reducing the sample **modulo 1**.
3. $\alpha.q$ is the standard deviation (not on $D_\alpha^{(1)}$).

The LWE Distribution $A_{\mathbf{s},\phi}$

The distribution $A_{\mathbf{s},\phi}$ over $\mathbb{Z}_q^n \times \mathbb{T}$ is generated by:

1. Choosing a vector $\mathbf{a} \in \mathbb{Z}_q^n$ **uniformly at random**.
2. Choosing an error term $e \in \mathbb{T}$ according to ϕ.
3. Outputting the pair:

$$\left(\mathbf{a}, \frac{\langle \mathbf{a}, \mathbf{s} \rangle}{q} + e\right) \in \mathbb{Z}_q^n \times \mathbb{T}$$

Note: All additions are performed in $\mathbb{T}$ (modulo 1).

Note: $r = \sigma\sqrt{2\pi}$ relates the Gaussian parameter to standard deviation.

The LWE Problem

Definition : The LWE Problem ($\mathrm{LWE}_{q,\phi}$)

Given access to any desired poly(n) number of samples from $A_{\mathbf{s},\phi}$ for some arbitrary $\mathbf{s} \in \mathbb{Z}_q^n$: **Goal:** Find $\mathbf{s}$ with overwhelming probability.

Remark: Worst-case to Average-case

- The definition is **worst-case** (true for any arbitrary $\mathbf{s}$).
- Using random self-reduction, this is equivalent to the **average-case** version where $\mathbf{s}$ is chosen uniformly at random.

Regev's Classical Reduction

The reduction R transforms a **worst-case Bounded-Distance Decoding (BDD)** problem into an LWE problem.

Proposition [Reg05, Lemma 3.4]

There is a classical PPT reduction $R^{W,D}(\mathbf{B}, r, \mathbf{x})$ that, given:

- Basis $\mathbf{B}$ of lattice Λ, real $r \geq \sqrt{2q} \cdot \eta_\epsilon(\Lambda^*)$
- Target point $\mathbf{x}$ within distance $\alpha q/(\sqrt{2}r) < \lambda_1(\Lambda)/2$
- **Oracle** W**:** Solves $\mathrm{LWE}_{q,\phi}$ using $\mathrm{poly}(n)$ samples generated by D
- **Oracle** D**:** Generates samples from discrete Gaussian $D_{\Lambda^*,r}$

Result: Finds the unique $\mathbf{v} \in \Lambda$ closest to $\mathbf{x}$.

- Succeeds with **overwhelming probability**.
- Relies on the trade-off between q and r to maintain the decoding radius.

Mechanism of the Reduction

How the reduction simulates the LWE distribution $A_{\mathbf{s},\phi}$:

The Transformation Steps

1. Let $\mathbf{s} = \mathbf{B}^{-1}\mathbf{v} \bmod q$ (the target coefficient vector).
2. Use Oracle D to obtain a dual sample $\mathbf{y} \leftarrow D_{\Lambda^*,r}$.
3. Let $\mathbf{a} = (\mathbf{B}^*)^{-1}\mathbf{y} = \mathbf{B}^t\mathbf{y} \bmod q$.
4. Output the LWE sample: $(\mathbf{a}, b = \langle \mathbf{y}, \mathbf{x} \rangle / q + e)$.

Why it works:

- $\mathbf{a}$ is essentially uniform because r is above the smoothing parameter.
- Because $\mathbf{x}$ and $\mathbf{v}$ are close, the inner product relates back to the secret:

$$\langle \mathbf{y}, \mathbf{x} \rangle \approx \langle \mathbf{y}, \mathbf{v} \rangle = \langle \mathbf{B}^t\mathbf{y}, \mathbf{B}^{-1}\mathbf{v} \rangle = \langle \mathbf{a}, \mathbf{s} \rangle \pmod q$$

Parameter Trade-offs

- **Decoding Radius:** Defined by $\frac{\alpha q}{\sqrt{2}r}$.
- **The q vs. r Trade-off:**
  - A larger r (wider Gaussian samples from Oracle D) requires a correspondingly larger *Modulus* q to keep the decoding radius stable.
- **The Role of α:** Typically the inverse of a small polynomial; its effect on the decoding radius is generally mild compared to q and r.

Conclusion of the Step

By invoking Oracle W on these simulated samples to find $\mathbf{s}$, the reduction can then recover the full vector $\mathbf{v}$ by iterating the procedure.

The Classical Hardness of LWE

This theorem establishes that LWE is at least as hard as classically solving specific versions of the Shortest Vector Problem (SVP).

Main Theorem/Reduction

Let $\alpha = \alpha(n) \in (0,1)$, $\gamma = \gamma(n) \geq n/(\alpha\sqrt{\log n})$. Let $\zeta = \zeta(n) \geq \gamma(n)$ and $q = q(n) \geq \zeta(n) \cdot \omega(\sqrt{\log n/n})$.

There exists a **probabilistic polynomial-time reduction** from solving:

$$\textbf{Worst-case GapSVP}_{\zeta,\gamma} \quad \Longrightarrow \quad \textbf{Average-case LWE}_{q,\Psi_\alpha}$$

- **Type:** Classical reduction (does not require a quantum computer).
- **Sample Complexity:** Uses only poly(n) samples.
- **Confidence:** Succeeds with overwhelming probability.

Parameter Implications and Generalizations

Remark : Modulus Size q

GapSVP$_{\zeta,\gamma}$ is potentially hard in the worst case whenever $\zeta > 2\gamma$. This allows for a choice of q as small as:

$$q > 2\gamma \cdot \omega(\sqrt{\log n/n}) = \omega(\sqrt{n}/\alpha)$$

Remark : Generalization to ℓ_p Norms

Using bounds from [Pei08a] on the smoothing parameter:

- Theorem 3.1 generalizes to GapSVP$_{\zeta,\gamma}$ in **any** ℓ_p norm for $2 < p \leq \infty$.
- The approximation factor $\gamma(n)$ remains essentially the same (up to $O(\log n)$ factors).

Key Takeaway: The security of LWE-based cryptosystems can be based on the hardness of solving geometric lattice problems in various norms, not just the standard ℓ_2 Euclidean norm.

Proof Architecture: From GapSVP to LWE

The reduction for the Main Theorem consists of two primary conceptual stages integrated into a single workflow.

Two-Step Reduction Path

$$\textbf{GapSVP} \xrightarrow{\text{[LM09]}} \textbf{BDD} \xrightarrow{\text{[Reg 05] Lemma 3.4}} \textbf{LWE}$$

The Perturbation Mechanism: Given a lattice Λ, the reduction performs the following:

1. **Perturb:** It picks a lattice point $\mathbf{v} \in \Lambda$ and perturbs it.
2. **Invoke:** It runs the reduction R (from Prop 3.4) on this perturbed point.
3. **Test:** It checks if R successfully recovers the original $\mathbf{v}$.

Case Analysis

- **If $\lambda_1(\Lambda)$ is Large:** By hypothesis, R is obliged to return the correct $\mathbf{v}$.
- **If $\lambda_1(\Lambda)$ is Small:** The perturbation *information-theoretically hides* $\mathbf{v}$ from R, forcing it to guess incorrectly.

Note: This behavior allows us to distinguish between the two cases, effectively playing a role similar to the prover in an interactive Arthur-Merlin proof [GG00].

Proof of Main Theorem: The Reduction

Input & Setup

An instance of $\text{GapSVP}_{\zeta,\gamma}$, i.e., a pair $(\mathbf{B}, d)$ where:

- $\min \|\widetilde{\mathbf{b}}_i\| \geq 1$
- $\lambda_1(\mathcal{L}(\mathbf{B})) \leq \zeta$ and $1 \leq d \leq \zeta/\gamma$
- Let $\Lambda = \mathcal{L}(\mathbf{B})$

The reduction runs the following procedure $N = poly(n)$ times:

1. **Sampling:** Choose $\mathbf{w}$ uniformly at random from the ball $d' \cdot \mathcal{B}_n$ where:

$$\|w\| < d' = d \cdot \sqrt{n/(4 \log n)} \quad \text{and let } \mathbf{x} = \mathbf{w} \pmod{\mathbf{B}}$$

2. **Oracle Invocation:** Invoke reduction R (from Prop 3.4) on $\mathbf{B}$ and $\mathbf{x}$ with parameter:

$$r = \frac{q \cdot \sqrt{2n}}{\gamma \cdot d}$$

where oracle D samples from $D_{\Lambda^*,r}$ using the reversed dual basis $\mathbf{D}$.

Proof of Main Theorem: Decision & Analysis

Decision Rule

Let $\mathbf{v}$ be the output (closest vector to $\mathbf{x}$) of R.

- If $\mathbf{v} \neq \mathbf{x} - \mathbf{w}$ in **any** of the N iterations $\implies$ **Accept**.
- Otherwise (i.e if $\mathbf{v} = \mathbf{x} - \mathbf{w}$ in **all** of the N iterations) $\implies$ **Reject**.

Parameter Analysis

Recall that $\max_i \|\widetilde{\mathbf{d}}_i\| = 1/\min_i \|\widetilde{\mathbf{b}}_i\| \leq 1$. The parameter r satisfies:

$$r = \frac{q \cdot \sqrt{2n}}{\gamma \cdot d} \geq \frac{q \cdot \sqrt{2n}}{\zeta} \geq \omega(\sqrt{\log n}) \geq \max_i \|\widetilde{\mathbf{d}}_i\| \omega(\sqrt{\log n})$$

- This hypothesis on d and q ensures the sampling algorithm [GPV08, Theorem 4.1] stays within **negligible statistical distance** of $D_{\Lambda^*,r}$.

Proof Analysis: The NO Instance

Case 1: $(\mathbf{B}, d)$ is a NO instance

In this case, $\lambda_1(\Lambda) > \gamma \cdot d$. By [MR07, Lemma 3.2]:

$$\eta_\epsilon(\Lambda^*) \leq \frac{\sqrt{n}}{\gamma \cdot d}$$

for $\epsilon(n) = 2^{-n} = \mathsf{negl}(n)$. Thus, $r \geq \sqrt{2q} \cdot \eta_\epsilon(\Lambda^*)$.

- Since $\mathbf{x} - \mathbf{w} \in \Lambda$, the distance from $\mathbf{x}$ to Λ is at most:

$$\|w\| < d' = d \cdot \sqrt{\frac{n}{4 \log n}} \leq \frac{\alpha \cdot \gamma \cdot d}{\sqrt{4n}} = \frac{\alpha q}{\sqrt{2}r}$$

- Moreover, $\frac{\lambda_1(\Lambda)}{2} > \frac{\gamma \cdot d}{2} > \frac{\gamma \cdot d \cdot \alpha}{2\sqrt{n}} = \frac{\alpha \cdot q}{\sqrt{2}r} > d' > \|w\|$.

Conclusion for NO Instance

The reduction R must return $\mathbf{v} = \mathbf{x} - \mathbf{w}$ in **each** iteration (with overwhelming probability). Therefore, the reduction **rejects** as desired.

Proof Completion: YES Instance Analysis

Statistical Distance Argument

In a YES instance ($\lambda_1(\Lambda) \leq d$), we consider the shift $\mathbf{w}' = \mathbf{z} + \mathbf{w}$ where $\mathbf{z} \in \Lambda$. By Lemma[GG00] and the data processing inequality for statistical distance:

Probability Inequalities

$$\begin{aligned}\Pr[R(\mathbf{x}) = \mathbf{x} - \mathbf{w}] &\leq 1 - \frac{1}{\text{poly}(n)} + \Pr[R(\mathbf{x}') = \mathbf{x}' - \mathbf{w}'] \\ &\leq 2 - \frac{1}{\text{poly}(n)} - \Pr[R(\mathbf{x}') = \mathbf{x}' - \mathbf{w}]\end{aligned}$$

- **Identity:** Note that $\mathbf{x}' = \mathbf{z} + \mathbf{w} \equiv \mathbf{w} \pmod{\mathbf{B}}$.
- **Result:** Since $\mathbf{x}'$ and $\mathbf{x}$ are distributed identically, we rearrange to find:

$$\Pr[R(\mathbf{x}) = \mathbf{x} - \mathbf{w}] \leq 1 - \frac{1}{\text{poly}(n)}$$

Conclusion for Yes Instance

For a sufficiently large $N = \text{poly}(n)$, there is at least one iteration where $\mathbf{v} \neq \mathbf{x} - \mathbf{w}$. Thus, the reduction **accepts** with high probability.

■

Thank You!

Classical Hardness of LWE
Questions? Discussion?

"The power of lattice cryptography lies in its worst-case security guarantees."

Presentation on Main Theorem and Background
Based on Regev et al.